

Power and Mounting



File: Power_and_Mounting.kicad_sch

Analog



File: Analog.kicad_sch

Digital

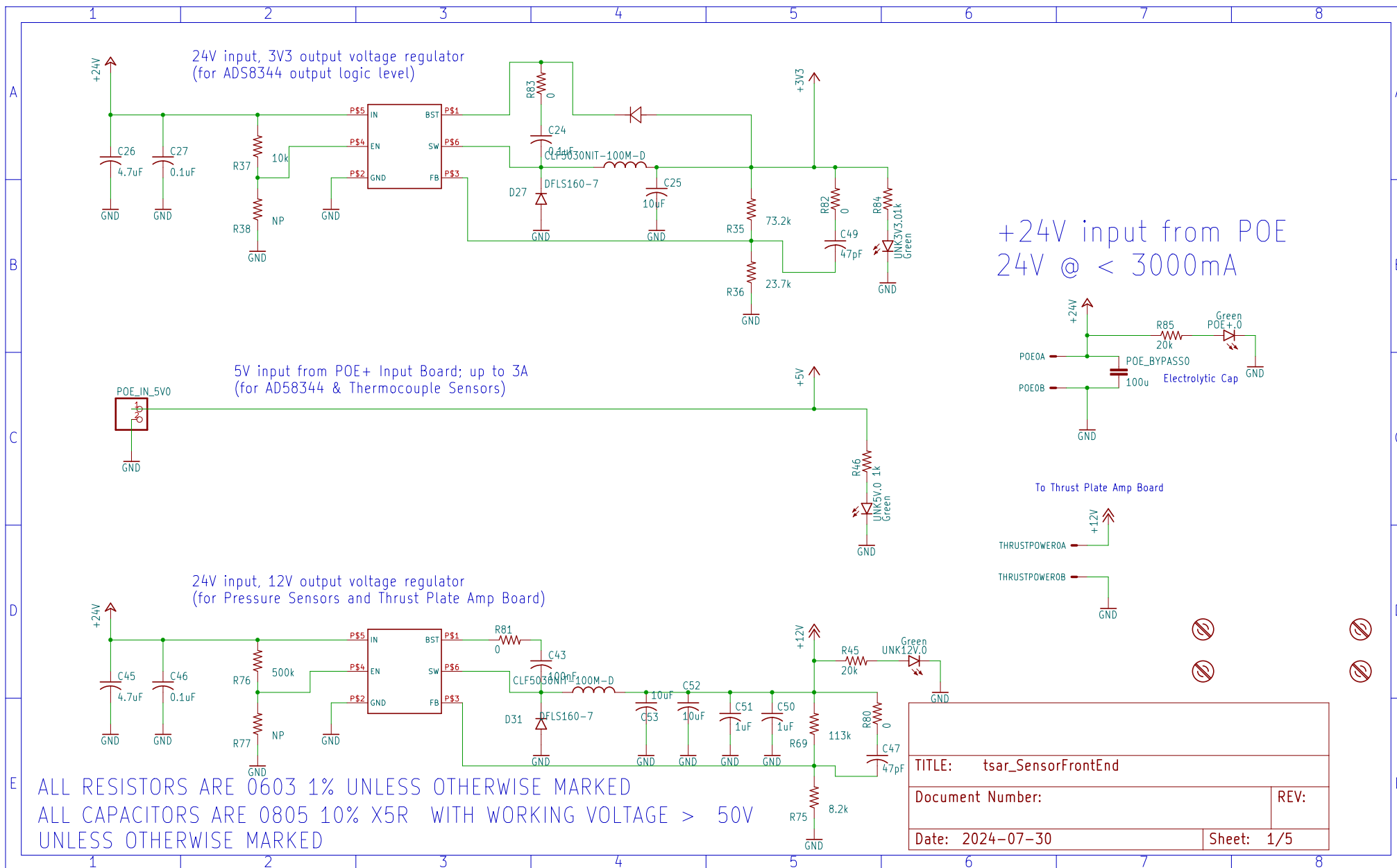


File: Digital.kicad_sch

Extra Sensors

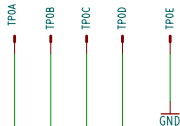


File: Extra_Sensors.kicad_sch



Thrust Plate Amp Connector

Thrust Plate Signals come from the Thrust Plate Amplifier Board. Sensors going into amp board are FC23 load cells.

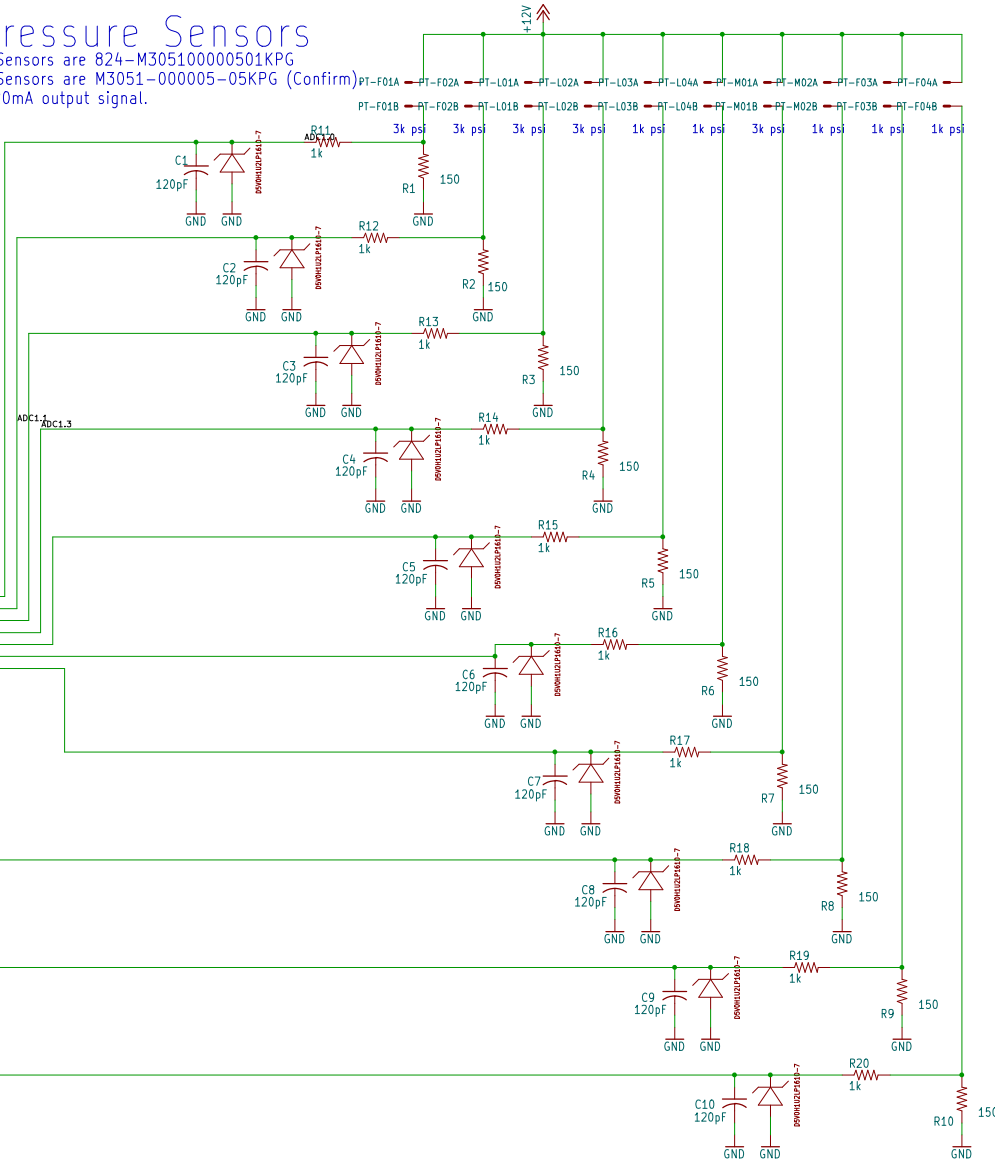
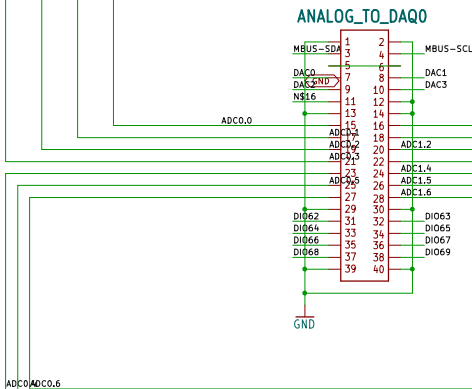


Pressure Sensors

1k Sensors are 824-M305100000501KPG

3k Sensors are M3051-000005-05KPG (Confirm)PT

4–20mA output signal.



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Temperature via Thermocouple

Temperature = $(V_{out} - 1.25) / 0.005$ V.
So, for example, if the voltage is 1.500V,
the temperature is $(1.5 - 1.25) / 0.005 = 50^{\circ}\text{C}$

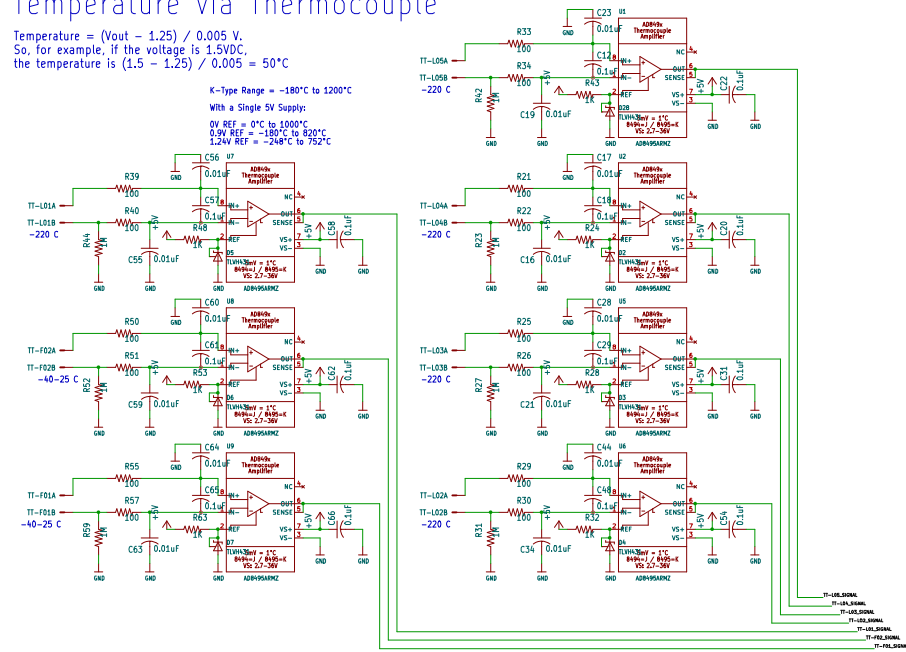
K-Type Range = -180°C to 1200°C

With a Single 5V Supply:

0V REF = 0°C to 1000°C

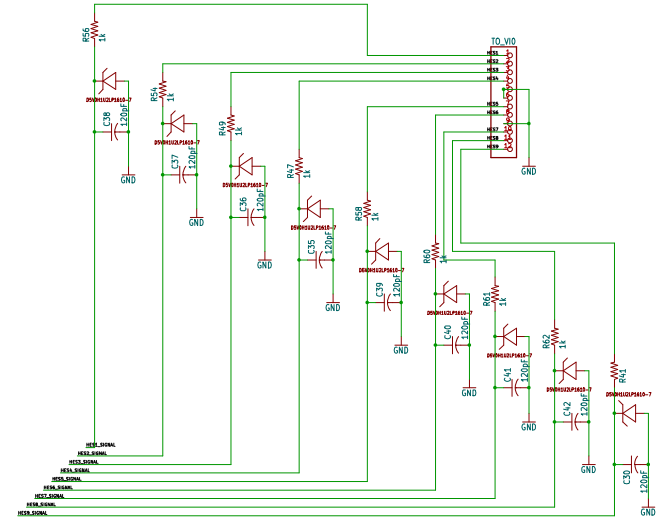
0.9V REF = -180°C to 820°C

1.24V REF = -240°C to 752°C



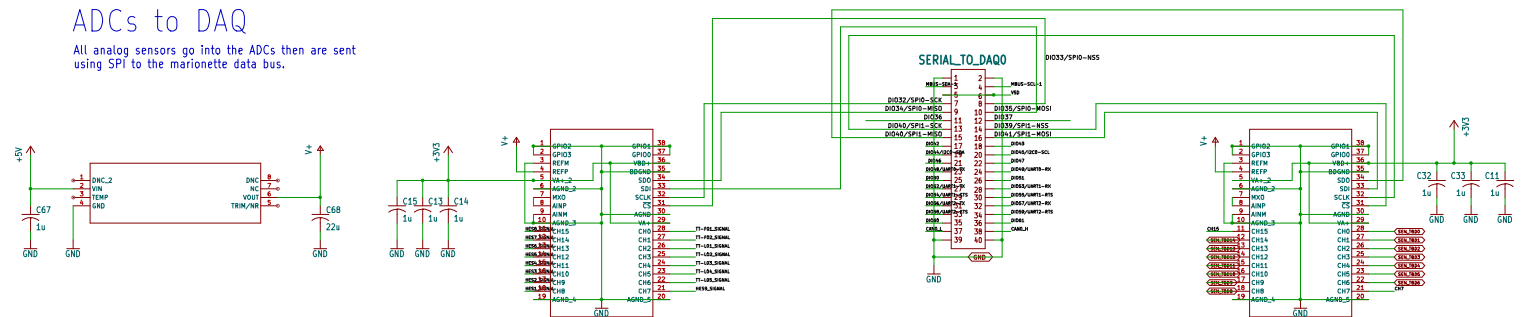
Valve Position Sensors

0V signal indicates the valves default position.
3.3V signal indicates the valves non-default position.



ADCs to DAQ

All analog sensors go into the ADCs then are sent
using SPI to the marionette data bus.



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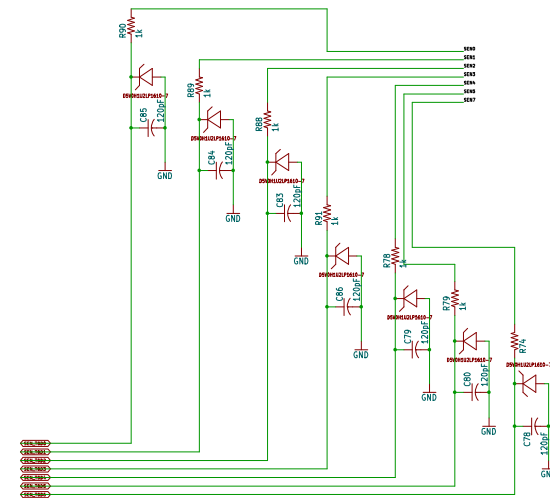
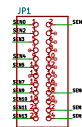
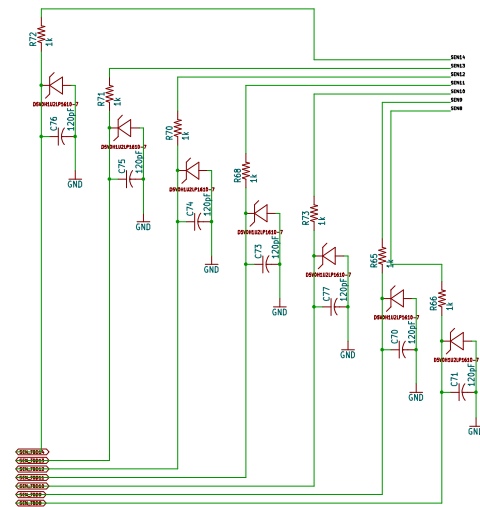
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Extra Sensor Inputs

Extra sensors with TVS protection.



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